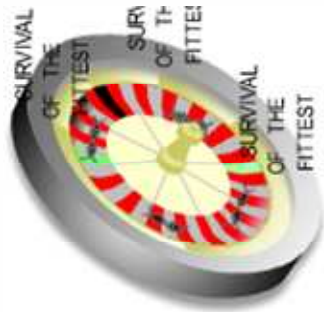




IEEE CONGRESS ON EVOLUTIONARY COMPUTATION

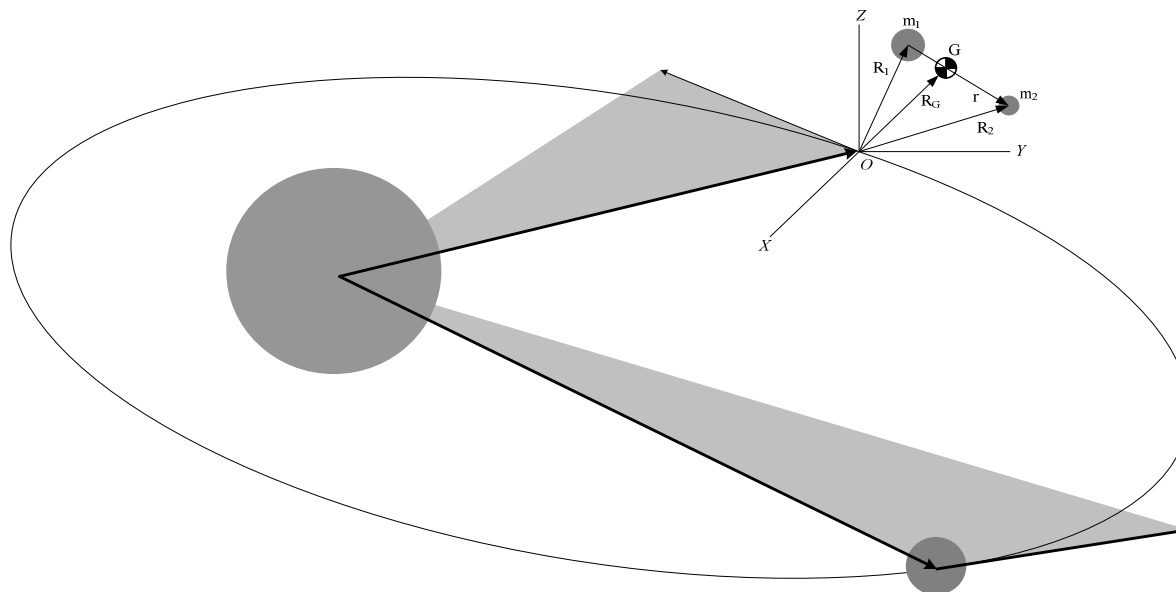
Singapore
September 25-28, 2007

www.cec2007.org



UNIVERSITY
of
GLASGOW

Multi-objective Optimisation of the Motorised Momentum Exchange Tether for Payload Orbital Transfer



Yi Chen

(yichen@mech.gla.ac.uk)

Matthew Cartmell

(m.cartmell@mech.gla.ac.uk)

*Department of Mechanical Engineering
University of Glasgow
<http://www.gla.ac.uk>*

26-Sep-2007

Contents

- ***Introduction***
- ***Concept of Tether Payload Transfer***
- ***Tether Stress***
- ***Payload Transfer Fitness Functions Definition***
- ***Simulations***
- ***Conclusions & Future Work***
- ***Acknowledgements***
- ***References***

Contents

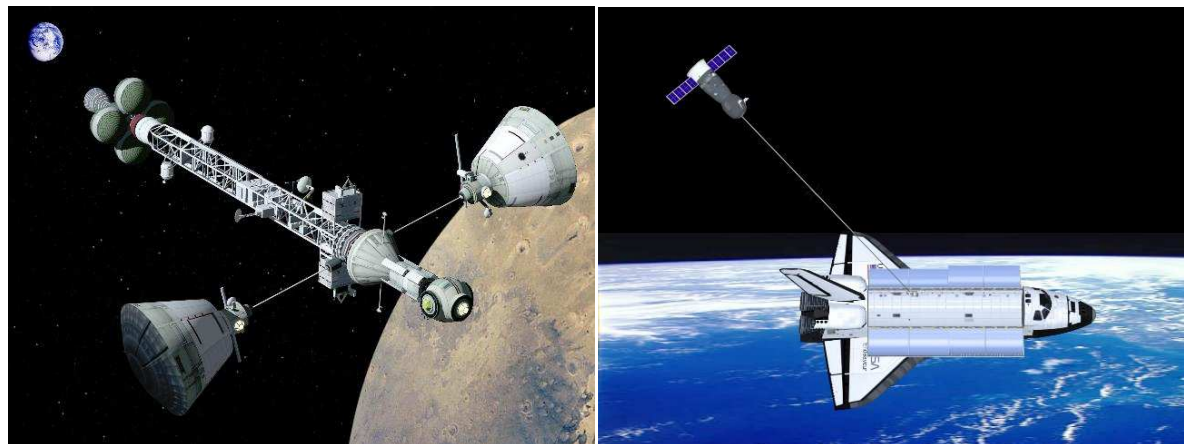
- ***Introduction***
- ***Concept of Tether Payload Transfer***
- ***Tether Stress***
- ***Payload Transfer Fitness Functions Definition***
- ***Simulations***
- ***Conclusions & Future Work***
- ***Acknowledgements***
- ***References***

What is Space Tether?

A **space tether** is a long **link** used to couple satellites, probes or spacecrafts to each other, or to other payloads.

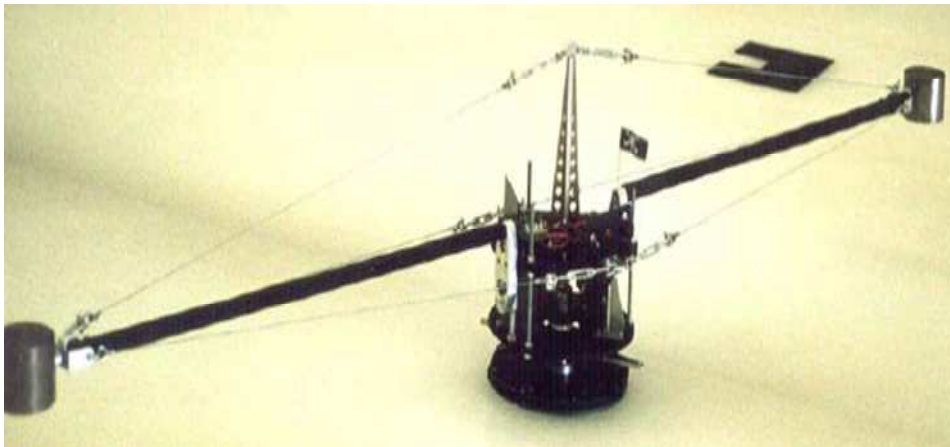
Space tethers are usually made of thin strands of high-strength fibres or conducting wires, which ranging from a few hundred metres to several kilometers, and have a **relatively** small diameter.

The tether can provide a mechanical **connection** between two space objects that enables the **transfer of energy and momentum** from one object to the other.

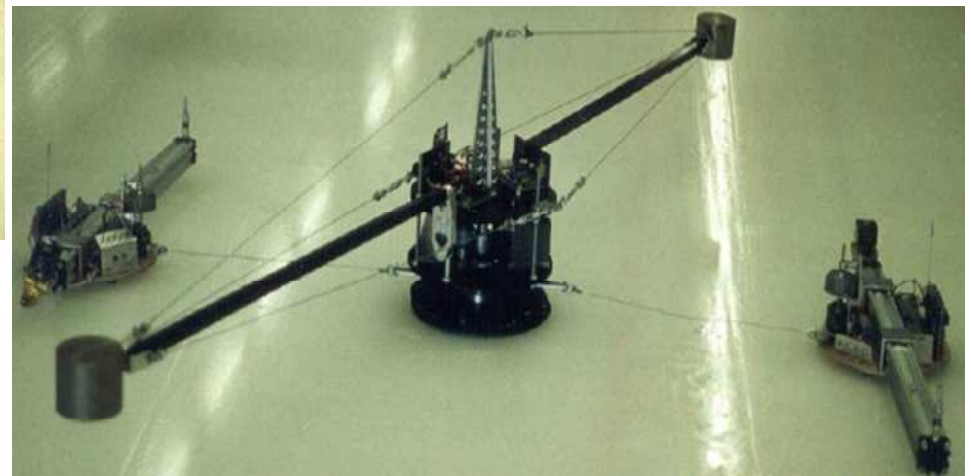


What is the Motorised Momentum Exchange Tether ?

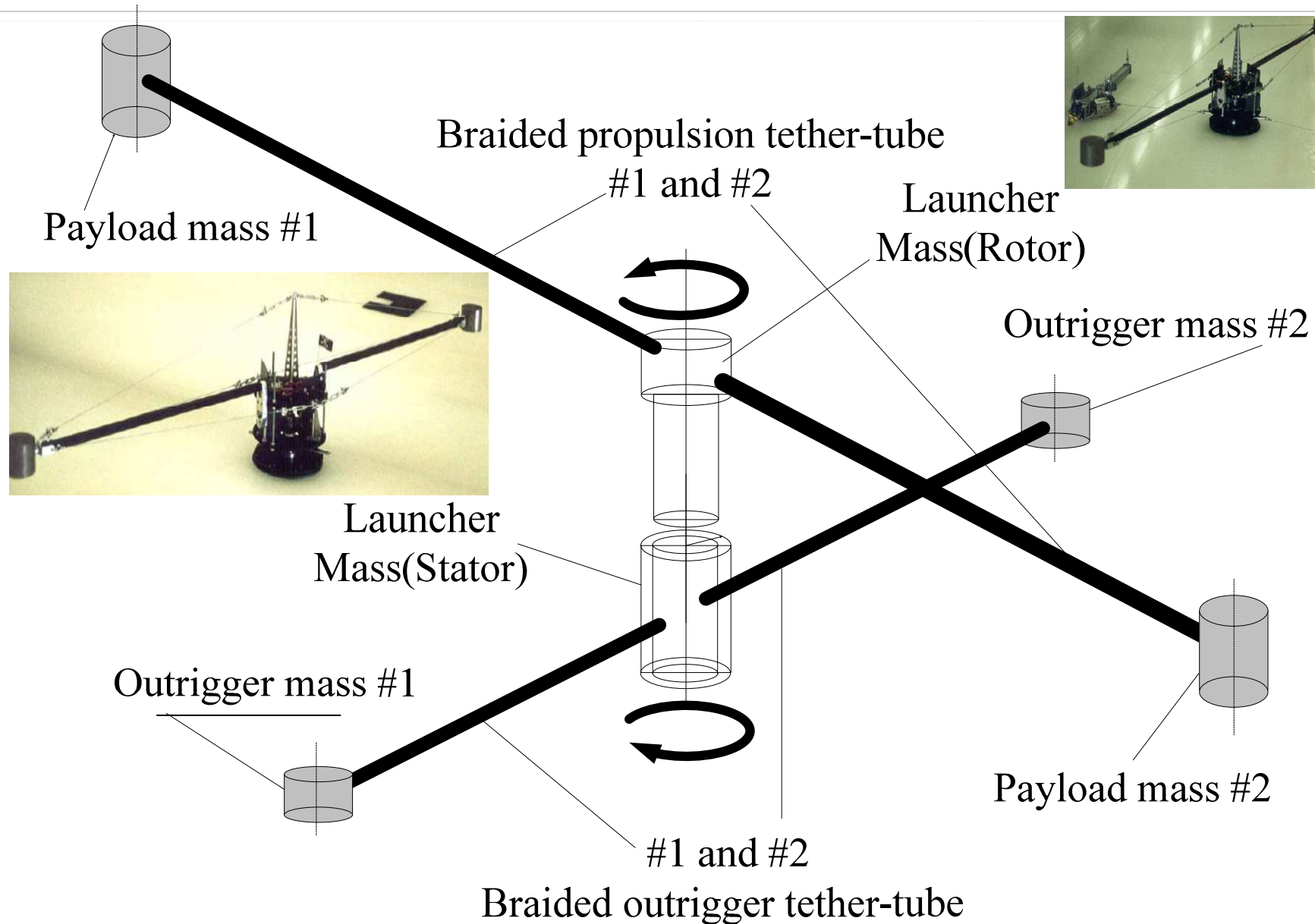
The motorised momentum exchange tether (**MMET**) was **first proposed** by Cartmell [1] and is based broadly on the conceptual schematic of the system to be shown. The model and concept were developed further, Ziegler and Cartmell [2], which led to proposals by Ziegler [3], who took physical angular co-ordinates for the generalised co-ordinates representing **spin** and **tilt**, and a further angular co-ordinate defining back-spin of the propulsion motor's stator components, known henceforth as the outrigger sub-system.



MMET Experiment on ice [10]



What is the Motorised Momentum Exchange Tether ?



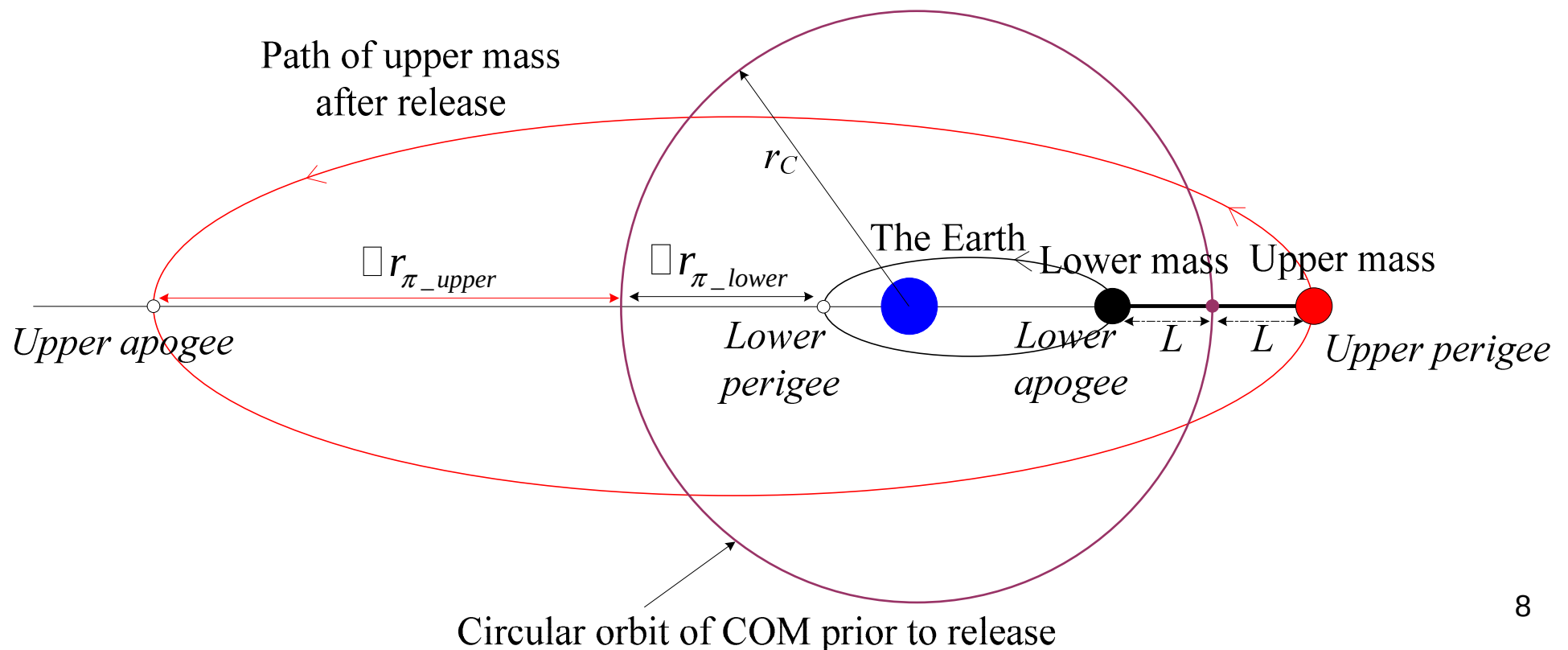
Contents

- ***Introduction***
- ***Concept of Tether Payload Transfer***
- ***Tether Stress***
- ***Payload Transfer Fitness Functions Definition***
- ***Simulations***
- ***Conclusions & Future Work***
- ***Acknowledgements***
- ***References***

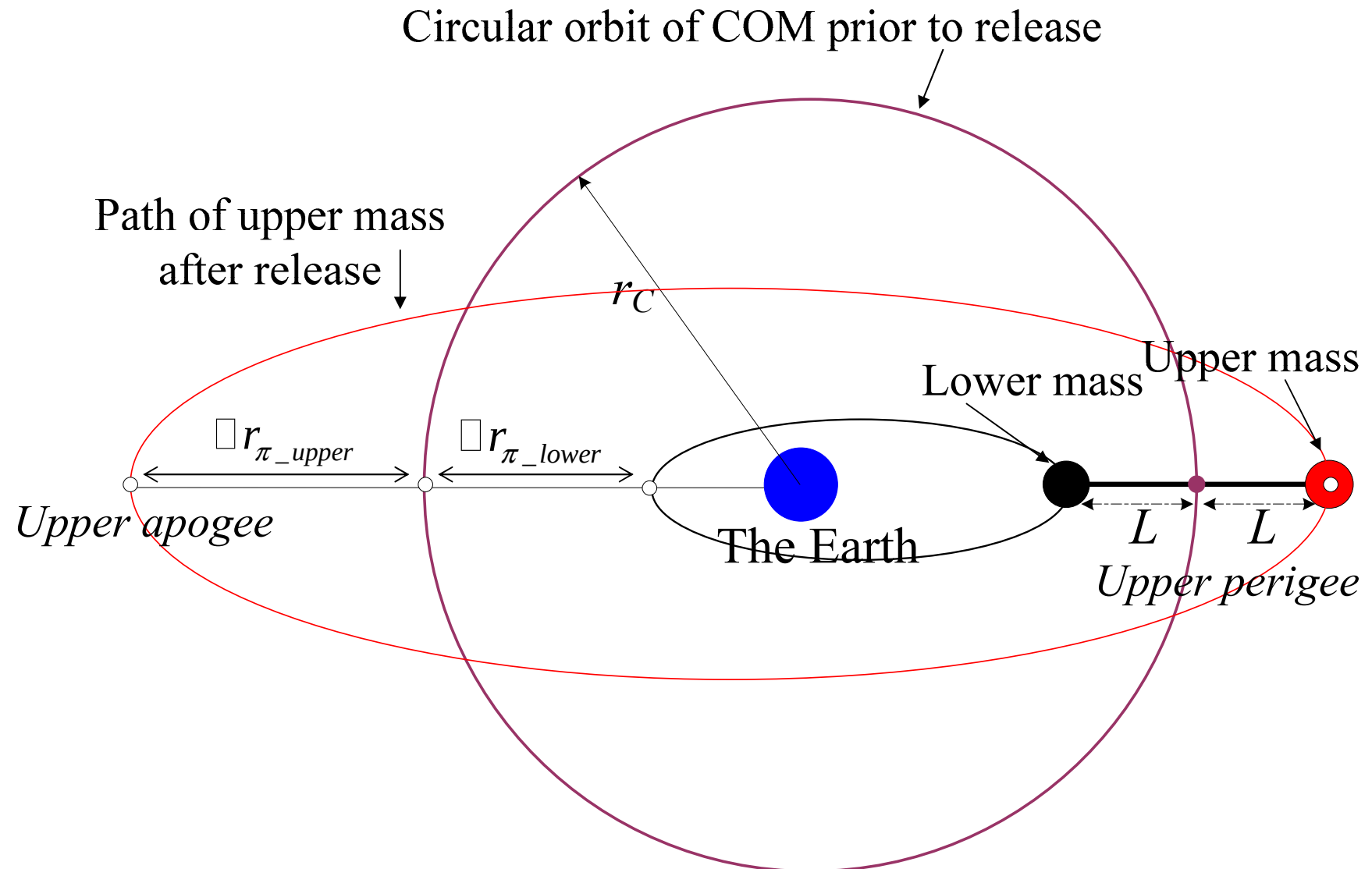
LEO Payload Transfer Paths

Perigee is the point at which an object in orbit around the Earth makes its **closest** approach to the Earth.

Apogee is the point at which an object in orbit around the Earth makes its **farthest** approach to the Earth.



Payload Efficiency Index Definition



Payload Efficiency Index Definition^[2]

upper payload efficiency index :

$$\frac{\Delta r_{\pi_upper}}{L} = 7 + \frac{30L}{r_c} + \frac{4\dot{\psi}}{\dot{\theta}} + \frac{2Lr_c^2\dot{\psi} (18\dot{\theta} + 5\dot{\psi})}{\mu}$$

lower payload efficiency index :

$$\frac{\Delta r_{\pi_lower}}{L} = -7 + \frac{30L}{r_c} - \frac{4\dot{\psi}}{\dot{\theta}} + \frac{2Lr_c^2\dot{\psi} (18\dot{\theta} + 5\dot{\psi})}{\mu}$$

TABLE I
TETHER PARAMETERS

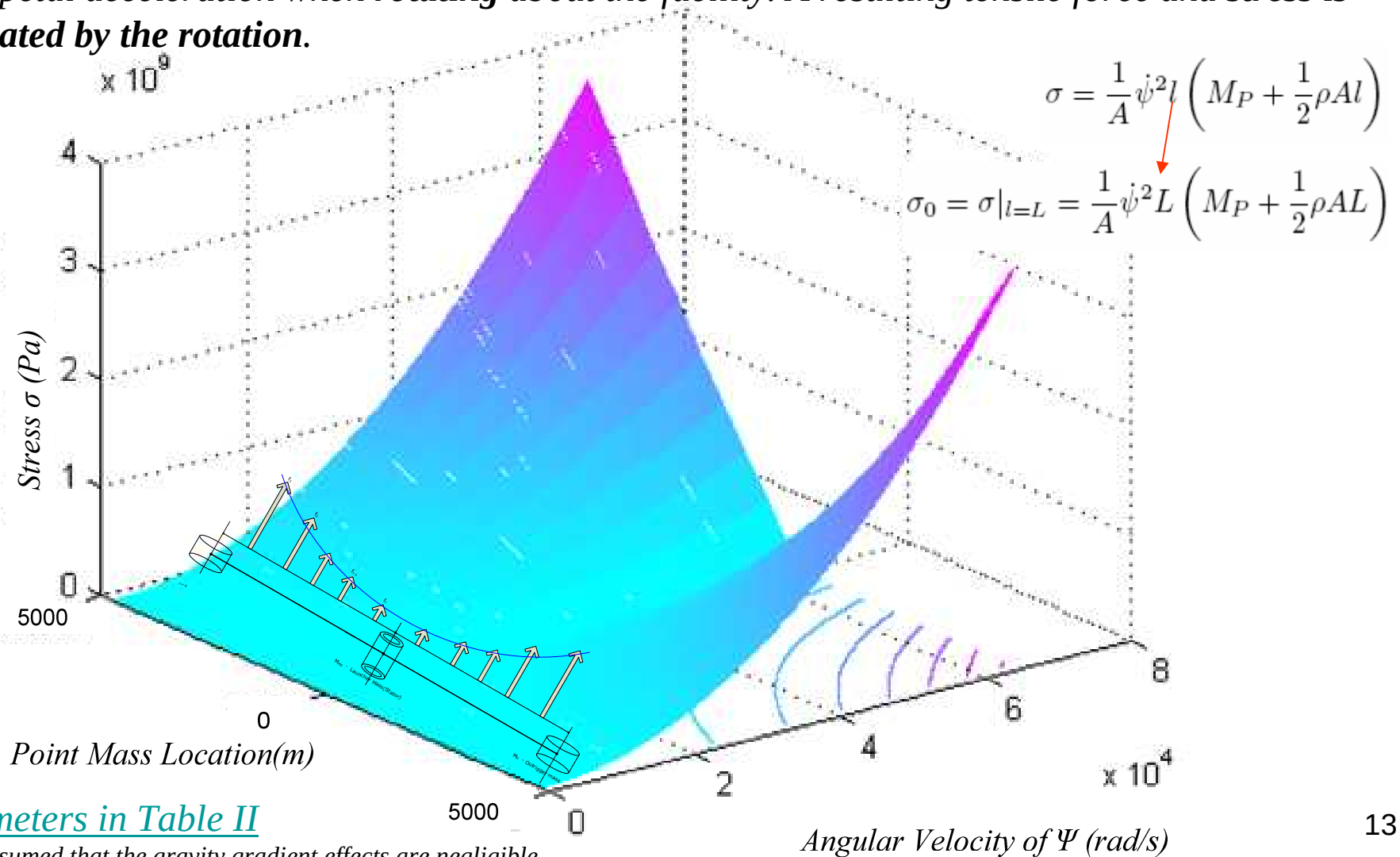
Symbol	Quantity	Value
A	tether cross-sectional area	$62.83 \times 10^{-6} m^2$
ρ	tether density	$970 kgm^{-3}$
T, U	kinetic and potential energy	J
M_M, M_P	motor and payload mass	$5000, 1000 kg$
Δr_π	tether performance index	m
$\Delta r_\pi / L$	efficiency index	
$\frac{\Delta r_{\pi_upper}}{L}$	upper payload efficiency index	
$\frac{\Delta r_{\pi_lower}}{L}$	lower payload efficiency index	
L	tether length	$50000 m$
l	tether length from COM a point along the tether	m
$\dot{\psi}$	angular pitch velocity	$0.01 rad/s$
$\dot{\theta}$	angular orbital velocity	$\sqrt{\mu/r_c} rad/s$
r_c	LEO circular orbit radius of COM at payload release	$6890 m$
σ_0	axial stress at the end of the tether where it connects to the payload	GPa
σ	stress	GPa
σ_{LIM}	stress limit	$5 GPa$
μ	gravitational constant	$3.9877848 \times 10^{14} m^3 s^{-2}$
e	eccentricity	0

Contents

- ***Introduction***
- ***Concept of Tether Payload Transfer***
- ***Tether Stress***
- ***Payload Transfer Fitness Functions Definition***
- ***Simulations***
- ***Conclusions & Future Work***
- ***Acknowledgements***
- ***References***

Tether Stress^[4]

The symmetrical motorised momentum exchange tether with a payload at one end will experience centripetal acceleration when **rotating** about the facility. A resulting tensile force and **stress is generated by the rotation**.



Parameters in Table II

If it is assumed that the gravity gradient effects are negligible

Angular Velocity of Ψ (rad/s)

Contents

- ***Introduction***
- ***Concept of Tether Payload Transfer***
- ***Tether Stress***
- ***Payload Transfer Fitness Functions Definition***
- ***Simulations***
- ***Conclusions & Future Work***
- ***Acknowledgements***
- ***References***

Payload Transfer Fitness Functions Definition

Tether Payload Transfer Problems:

Max Fitness Functions:

A. *Tether strength*

$$\text{Min } y_1 = \sigma_0$$

$$\longrightarrow f_1(\dot{\psi}, L, A) = \sigma_{LIM} - \sigma_0$$

B. *Upper payload transfer*

$$\text{Max } y_2 = \frac{\Delta r_{\pi_upper}}{L}$$

$$\longrightarrow f_2(\dot{\psi}, L) = \frac{\Delta r_{\pi_upper}}{L}$$

C. *Lower payload transfer*

$$\text{Min } y_3 = \frac{\Delta r_{\pi_lower}}{L}$$

$$\longrightarrow f_3(\dot{\psi}, L) = \frac{1}{\frac{\Delta r_{\pi_lower}}{L} + \varepsilon}$$

Subject to: $\sigma_0 \preceq \sigma_{LIM}$

where, ε is a small value parameter,
which helps to avoid
 $f_3 \rightarrow \infty$, and $\varepsilon \rightarrow 0$.

Parameters in Table II

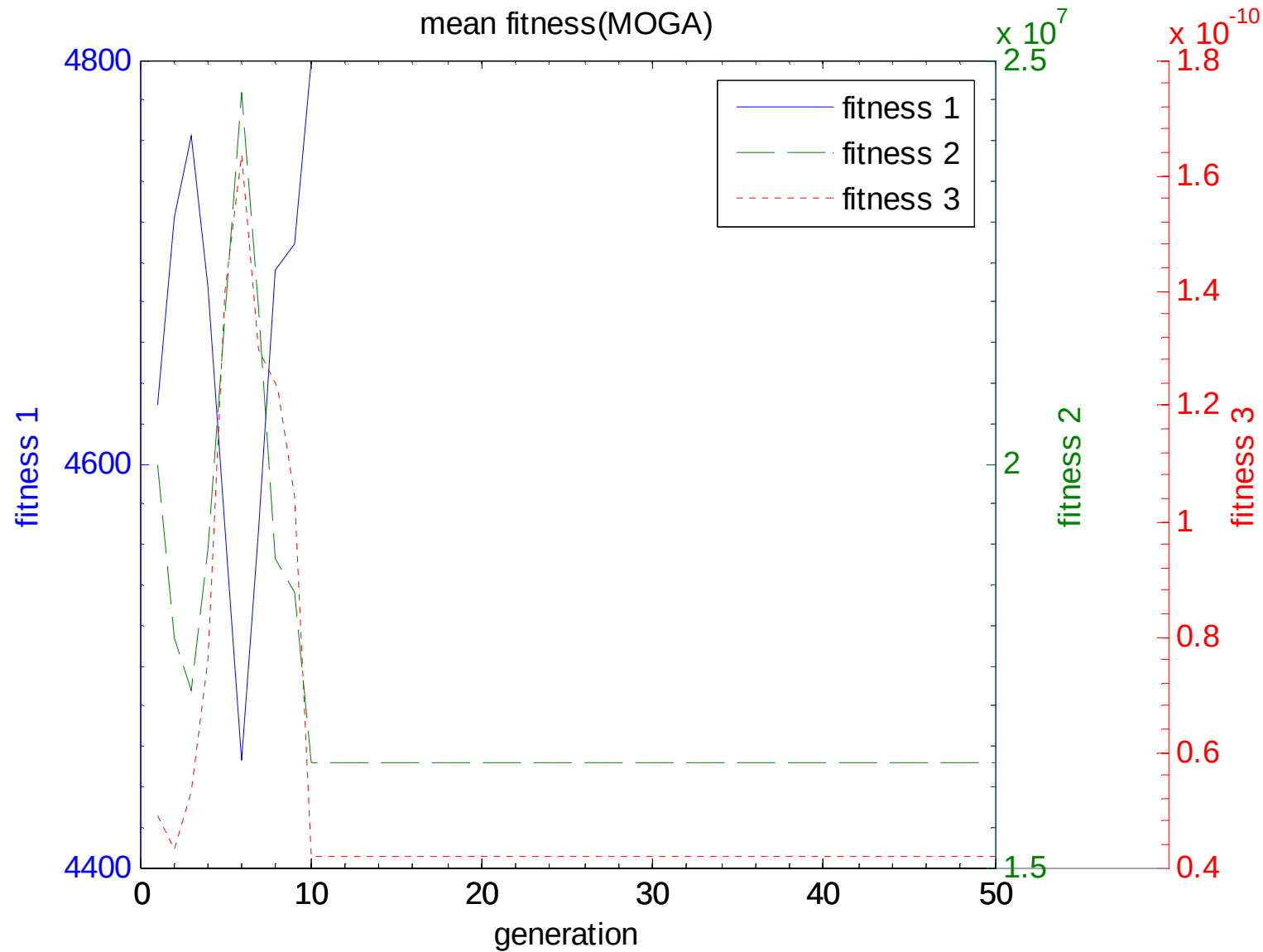
Contents

- ***Introduction***
- ***Concept of Tether Payload Transfer***
- ***Tether Stress***
- ***Payload Transfer Fitness Functions Definition***
- ***Simulations***
- ***Conclusions & Future Work***
- ***Acknowledgements***
- ***References***

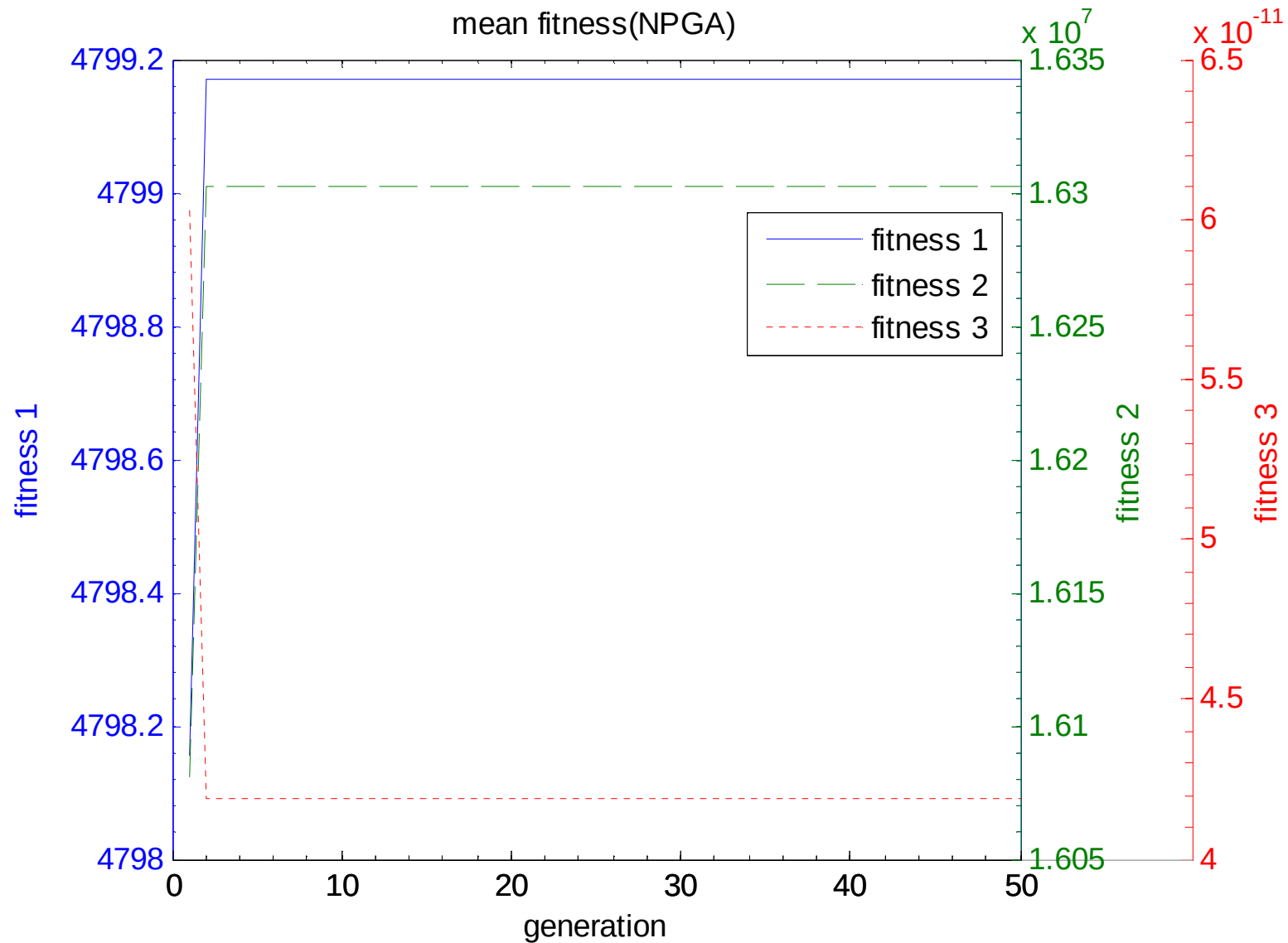
TABLE II
GAS PARAMETER SETTING

Symbol	Quantity	Value
	max generations crossover probability mutation probability population selection operator crossover operator mutation operator encoding method	50 0.8 0.001 50 tournament single point single point binary
x_1	ψ	$[1 \times 10^{-5}, 1]$
x_2	L	$[1 \times 10^3, 2.5 \times 10^5]$
x_3	A	$[5 \times 10^{-6}, 5 \times 10^{-1}]$
y_1	tether stress object function 	
y_2	upper efficiency index object function 	
y_3	lower efficiency index object function	
f_1	fitness function 1	
f_2	fitness function 2 	
f_3	fitness function 3 	
$\varepsilon \rightarrow$	a small value parameter, which helps to avoid $f_3 \rightarrow \infty$.	
		eps:the floating point relative accuracy

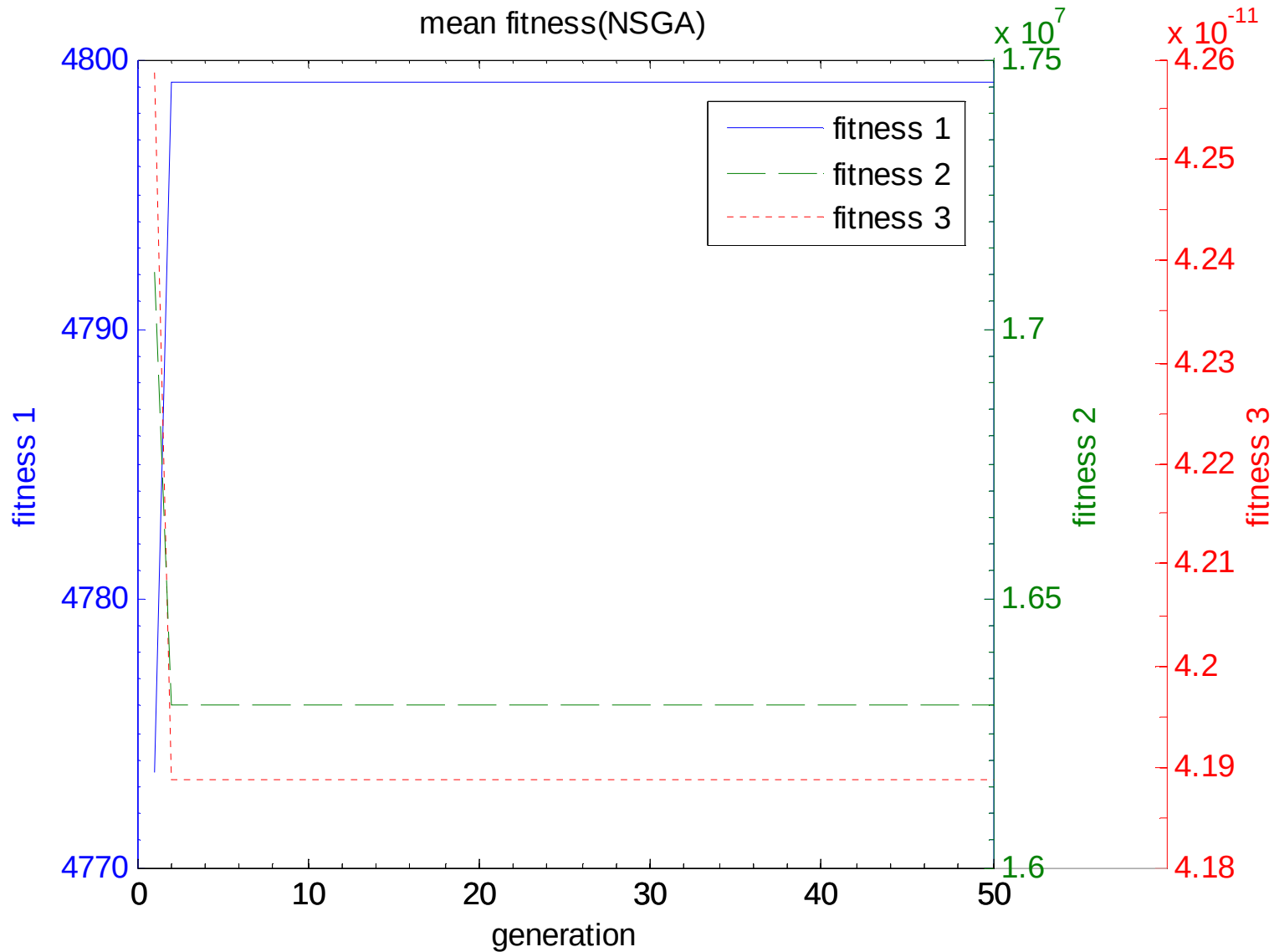
Fitness values of MOGA



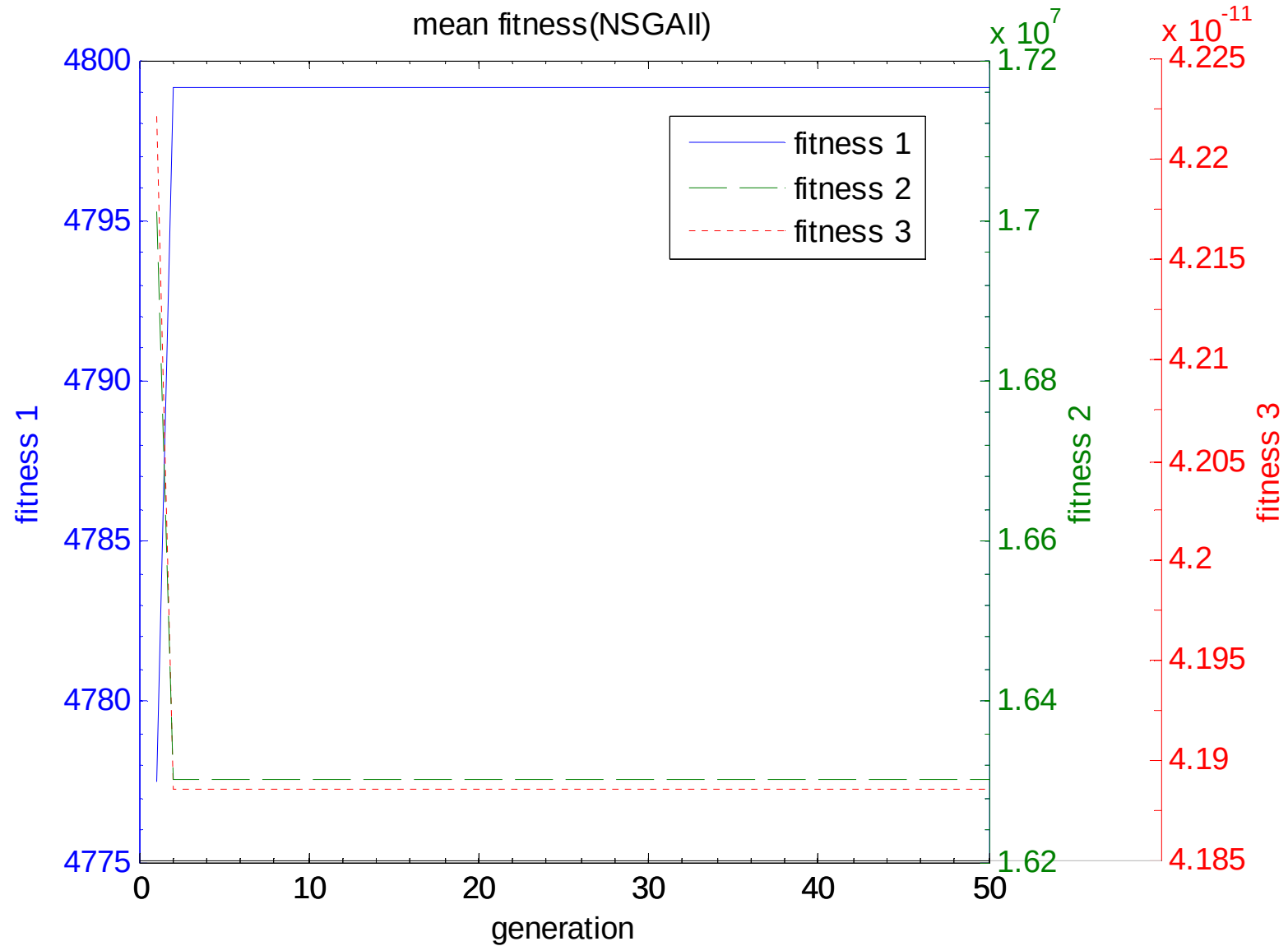
Fitness values of NPGA



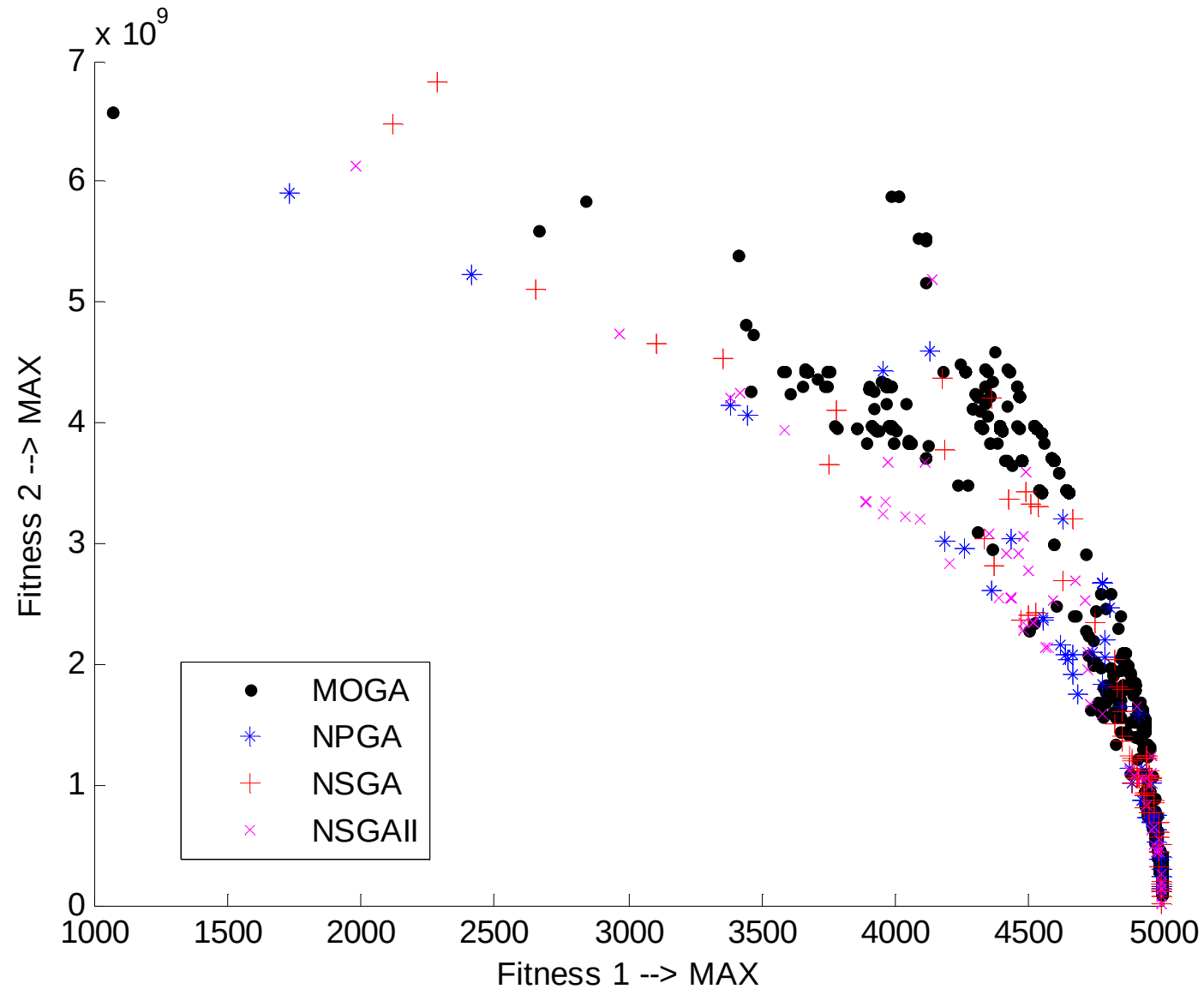
Fitness values of NSGA



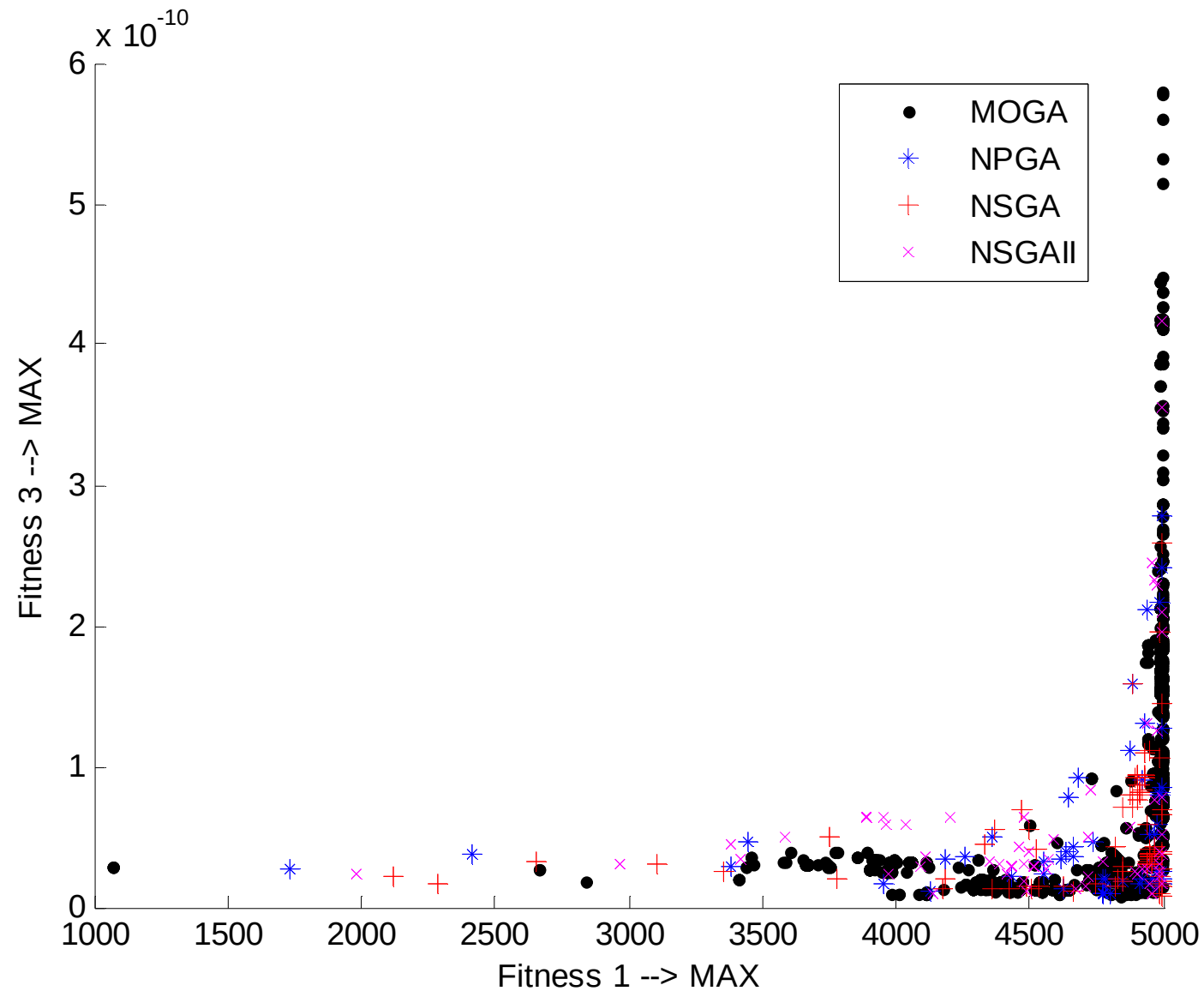
Fitness values of NSGAI



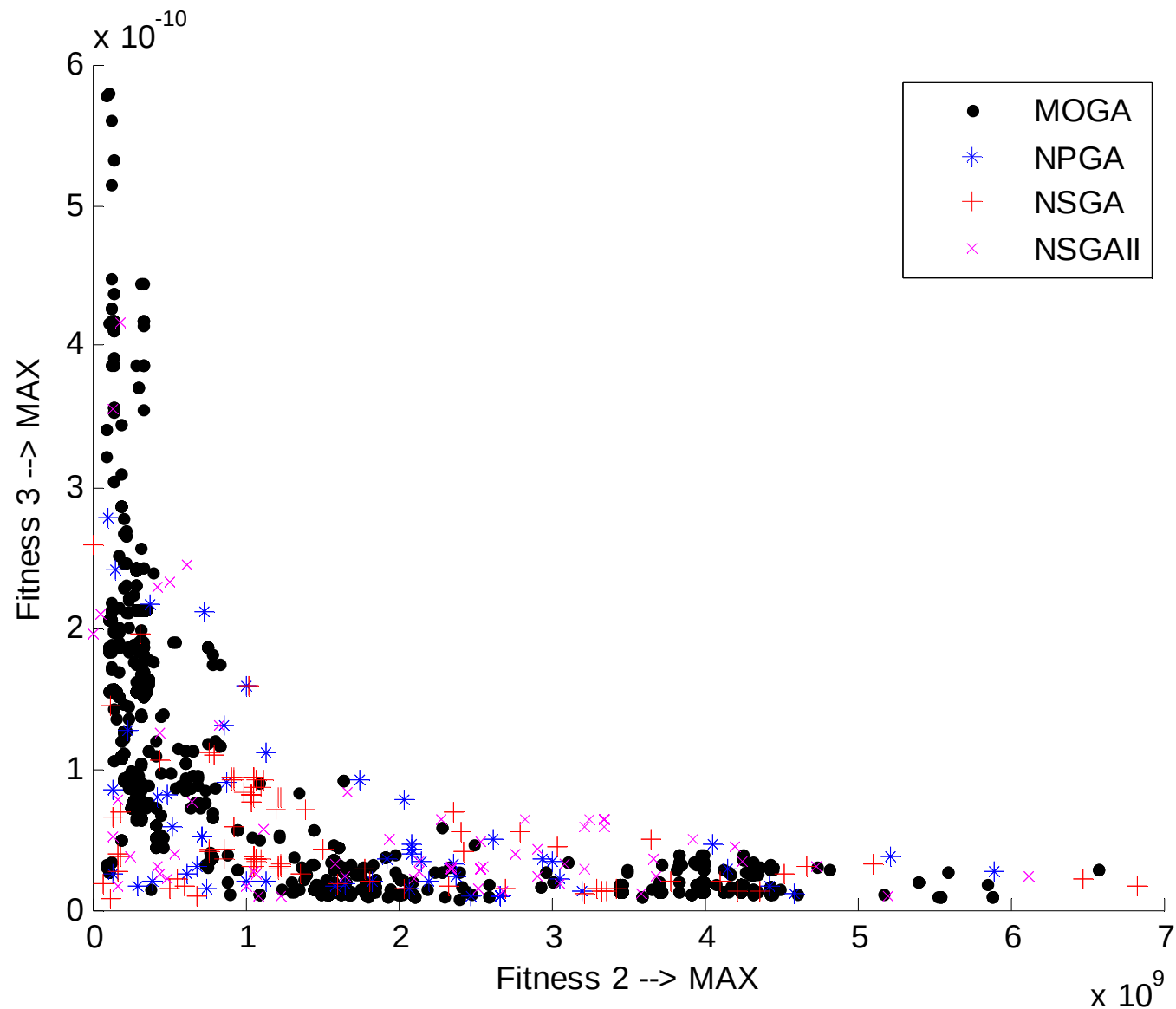
Pareto Front $\rightarrow$ fitness 1 and fitness 2



Pareto Front → fitness 1 and fitness 3



Pareto Front $\rightarrow$ fitness 2 and fitness 3



Contents

- ***Introduction***
- ***Concept of Tether Payload Transfer***
- ***Tether Stress***
- ***Payload Transfer Fitness Functions Definition***
- ***Simulations***
- ***Conclusions & Future Work***
- ***Acknowledgements***
- ***References***

Conclusion & Future Work

Conclusions:

- The 4 multi-objective methods of MOGA, NPGA, NSGA and NSGAI showed **flexible optimisation ability for the MMET practical objectives**, and the Pareto dataset suggests a series of solutions for the MMET, designing for the strength criterion and payload transfer performance.
- The GAs simulation helps to validate the current MMET payload transfer design
- more specific or **practical conditions** for each MMET application.

Future work:

- more **multi-objective GAs methods** will be applied into the current MMET payload transfer study.
- some optimisation studies for **multiple orbit MMET payload transfers**.
- intelligent real-time **controller design** for MMET payload transfer or navigation.
- dynamical system **modelling** for MMET.

Contents

- ***Introduction***
- ***Concept of Tether Payload Transfer***
- ***Tether Stress***
- ***Payload Transfer Fitness Functions Definition***
- ***Simulations***
- ***Conclusions & Future Work***
- ***Acknowledgements***
- ***References***

Acknowledgements

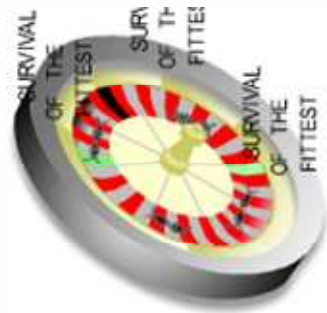
*The authors would like to acknowledge the support provided to the first author by the **Overseas Research Students Awards Scheme** and the **Scholarship** awarded by the University of Glasgow's **Faculty of Engineering**.*

Contents

- ***Introduction***
- ***Concept of Tether Payload Transfer***
- ***Tether Stress***
- ***Payload Transfer Fitness Functions Definition***
- ***Simulations***
- ***Conclusions & Future Work***
- ***Acknowledgements***
- ***References***

References

- [1] M. P. Cartmell, *Generating Velocity Increments by means of a Spinning Motorised Tether*, 34th AIAA/ASME/SAE/ASEE Joint propulsion Conference and Exhibit, Cleveland Conference Centre, Cleveland, Ohio, USA, AIAA-98-3739, 1998.
- [2] Ziegler, S.W., and Cartmell, M.P., *Using Motorised Tethers for Payload Orbital Transfer*, Journal of Spacecraft and Rockets, 38 (6), November / December, pp. 904-913, 2001.
- [3] Ziegler, S.W., *The Rigid-body Dynamics of Tethers in Space*, PhD Dissertation, Department of Mechanical Engineering, University of Glasgow, 2003.
- [4] Yi Chen, Matthew P. Cartmell., *Dynamical Modelling of The Motorised Momentum Exchange Tether Incorporating Axial Elastic Effects*, Advanced Problems in Mechanics -, 20-28 June, Russian Academy of Sciences, St. Petersburg, Russia, 2007.
- [5] Holland J H., *Adaptation in Natural and Artificial Systems*, The University of Michigan Press, 1975.
- [6] Fonseca C.M., Fleming P.J., *Genetic algorithms for multi-objective optimisation: Formulation, discussion and generalization*, In: S. Forrest (ed.), Genetic algorithms: Proceedings of the Fifth International Conference, Morgan Kaufmann, San Mateo, CA, 141-153, 1993.
Available:<http://citeseer.ist.psu.edu/fonseca93genetic.html>
- [7] Horn, J. Nafpliotis, N. Goldberg, D.E., *A niched Pareto genetic algorithm for multiobjective optimisation*, Evolutionary Computation, IEEE World Congress on Computational Intelligence, Proceedings of the First IEEE Conference on 27-29 Jun, vol. 1, pp. 82-87, 1994.
- [8] Srinivas N, Deb K., *Multiobjective optimisation using nondominated sorting in genetic algorithms*, Evolutionary Computation, vol. 2(3):pp. 221-248, 1994.
- [9] Deb, K., A. Pratap, S. Agarwal and T. Meyarivan, *A Fast and Elitist Multiobjective Genetic Algorithm: NSGAII*, IEEE Trans. on Evolutionary Comput., vol. 6, no. 2, pp. 182-197, 2002.
- [10] Cartmell, M.P. and Ziegler, S.W., *Experimental Scale Model Testing of a Motorised Momentum Exchange Propulsion Tether*, 37th AIAA/ASME/SEA/ASEE Joint Propulsion Conference and Exhibition, Salt Lake City, Utah, USA, AIAA 2001-3914, 8-11 July 2001.



IEEE CONGRESS ON EVOLUTIONARY COMPUTATION

Singapore
September 25-28, 2007

www.cec2007.org



UNIVERSITY
of
GLASGOW

Multi-objective Optimisation for the Motorised Momentum Exchange Tether Applied to Payload Orbital Transfer

Yi Chen

(yichen@mech.gla.ac.uk)

Matthew Cartmell

(m.cartmell@mech.gla.ac.uk)

Thanks

Department of Mechanical Engineering
University of Glasgow
<http://www.gla.ac.uk>

Any Questions?

End

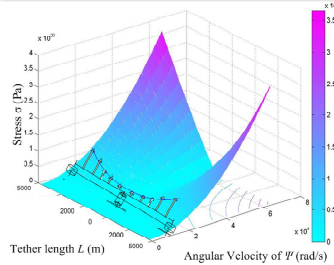


Multi-objective Optimisation of the Motorised Momentum Exchange Tether for Payload Orbital Transfer

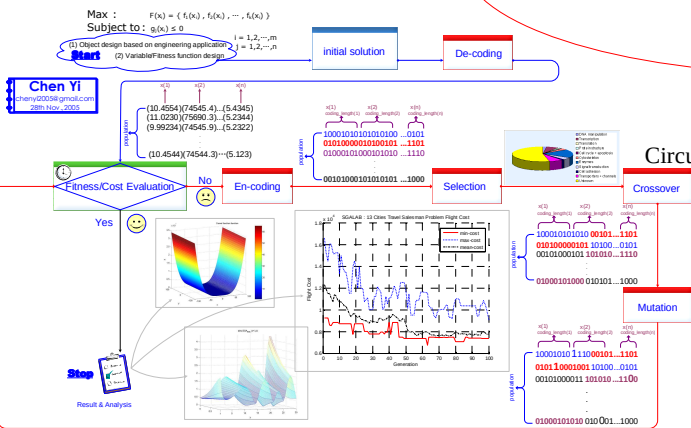
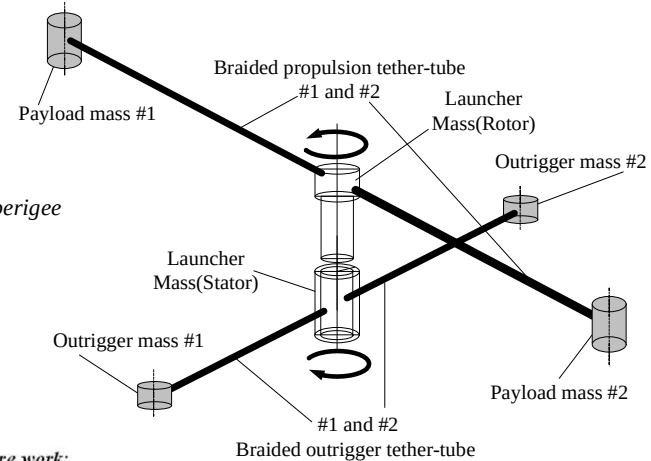
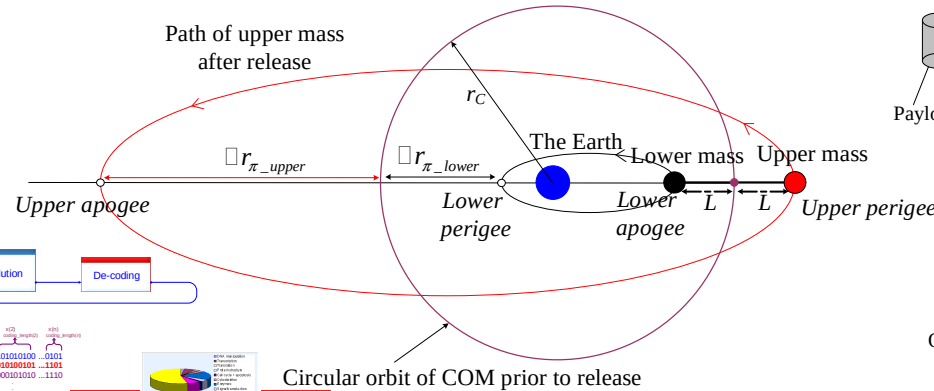
Yi Chen
(yichen@mech.gla.ac.uk)

Matthew Cartmell
(m.cartmell@mech.gla.ac.uk)

Department of Mechanical Engineering
University of Glasgow
http://www.gla.ac.uk



The motorized momentum exchange tether (MMET) was first proposed by Cartmell, then discussed further by Ziegler and Cartmell, and a more developed idealized MMET of spinning system with point masses was proposed by Ziegler, which motorized by a motor driver and took physical angular coordinates for the generalised coordinates representing spin and then tilt, together with an angular coordinate for circular orbital motion and a further angular co-ordinate defining backspin of the propulsion motor's stator components. This poster shows the last researches on MMET with axial elastic effects.



Conclusions:

- The 4 multi-objective methods of MOGA, NPGA, NSGA and NSGAII showed **flexible optimisation ability for the MMET practical objectives**, and the Pareto dataset suggests a series of solutions for the MMET, designing for the strength criterion and payload transfer performance.
- The GAs simulation helps to **validate** the current MMET payload transfer design
- more specific or **practical conditions** for each MMET application.

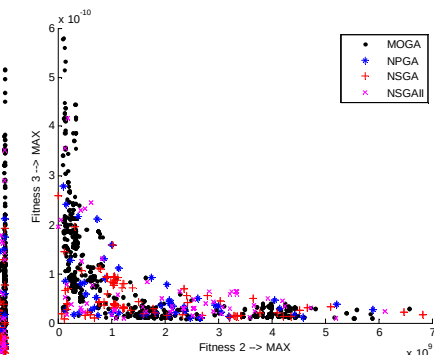
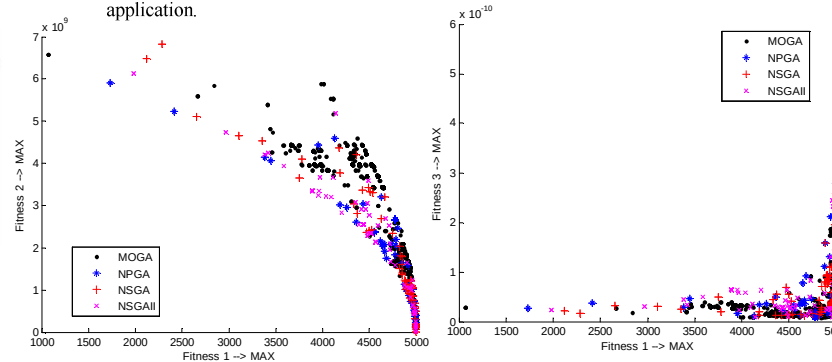
Future work:

- more **multi-objective GAs methods** will be applied into the current MMET payload transfer study.
- some optimisation studies for **multiple orbit MMET payload transfers**.
- intelligent real-time **controller design** for MMET payload transfer or navigation.
- dynamical system **modelling** for MMET.

Tether Payload Transfer Problems:

Max Fitness Functions:

- Tether strength**
Min : $y_1 = \sigma_0$ $\longrightarrow f_1(\psi, L, A) = \sigma_{LIM} - \sigma_0$
 - Upper payload transfer**
Max : $y_2 = \frac{\Delta r_{\pi_upper}}{L}$ $\longrightarrow f_2(\psi, L) = \frac{\Delta r_{\pi_upper}}{L}$
 - Lower payload transfer**
Min : $y_3 = \frac{\Delta r_{\pi_lower}}{L}$ $\longrightarrow f_3(\psi, L) = \frac{1}{\Delta r_{\pi_lower} + \varepsilon}$
- where, ε is a small value parameter, which helps to avoid $f_3 \rightarrow \infty$, and $\varepsilon \rightarrow 0$.
- Subject to: $\sigma_0 \leq \sigma_{LIM}$



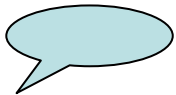
Acknowledgements
The authors would like to acknowledge the funding support of the Overseas Research Students Awards Scheme and the Faculty of Engineering Scholarship of University of Glasgow.

FAQ 1 - eps

eps is the minimal distance between any two numbers in Matlab if you're using Matlab's default double-precision.

This is an inherent effect from the way that doubles are represented in a computer according to the IEEE 754 standard.

See here: http://en.wikipedia.org/wiki/Floating_point#Behavior_of_computer_arithmetic



FAQ 2 – What is GAs?

Genetic Algorithms (GAs) are global, parallel, stochastic search methods, founded on Darwinian evolutionary principles, by Holland, 1975 [5].

GAs work with a population of potential solutions to a problem, and each individual within the population represents a specific solution to the problem, which is generally expressed as some form of genetic code. Since then, GAs have been frequently applied as optimisers for many engineering applications.

<http://www.mathworks.com/matlabcentral/fileexchange/loadFile.do?objectId=5882&objectType=file>

FAQ 2 – What is GAs?

Max : $F(x_i) = \{ f_1(x_i), f_2(x_i), \dots, f_k(x_i) \}$
 Subject to: $g_j(x_i) \leq 0$

- (1) Object design based on engineering application
Start (2) Variable/Fitness function design

initial solution

De-coding

Chen Yi

chenyi2005@gmail.com
 28th Nov., 2005

population
 $x(1)$ $x(2)$ $x(n)$
 (10.4554)(74545.4)...(5.4345)
 (11.0230)(75690.3)...(5.2344)
 (9.99234)(74545.9)...(5.2322)
 ...
 (10.4544)(74544.3)...(5.123)

population
 $x(1)$ $x(2)$ $x(n)$
 coding_length(1) coding_length(2) coding_length(n)
 10001010101010100...0101
 01010000010100101...1101
 01000101000101010...1110
 ...
 00101000101010101...1000

Fitness/Cost Evaluation



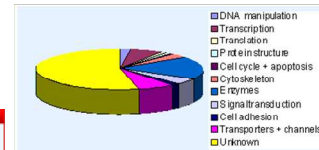
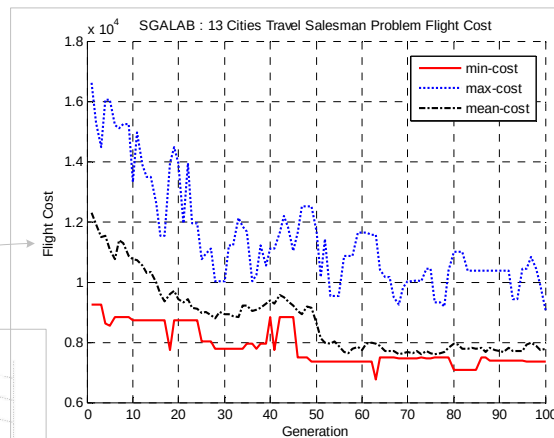
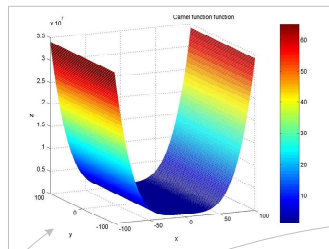
No

En-coding

Selection

Crossover

Yes

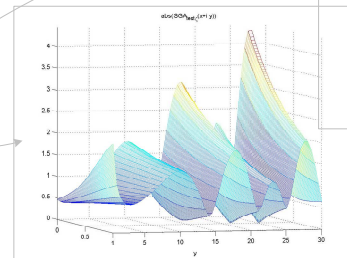


Mutation

population
 $x(1)$ $x(2)$ $x(n)$
 coding_length(1) coding_length(2) coding_length(n)
 10001010101000101...1101
 01010000010110100...0101
 00101000101101010...1110
 ...
 0100010100010101...1000
 ...
 10001010111000101...1101
 01011000100110100...0101
 00101000011101010...1100
 ...
 01000101010010001...1000

Stop

Result & Analysis



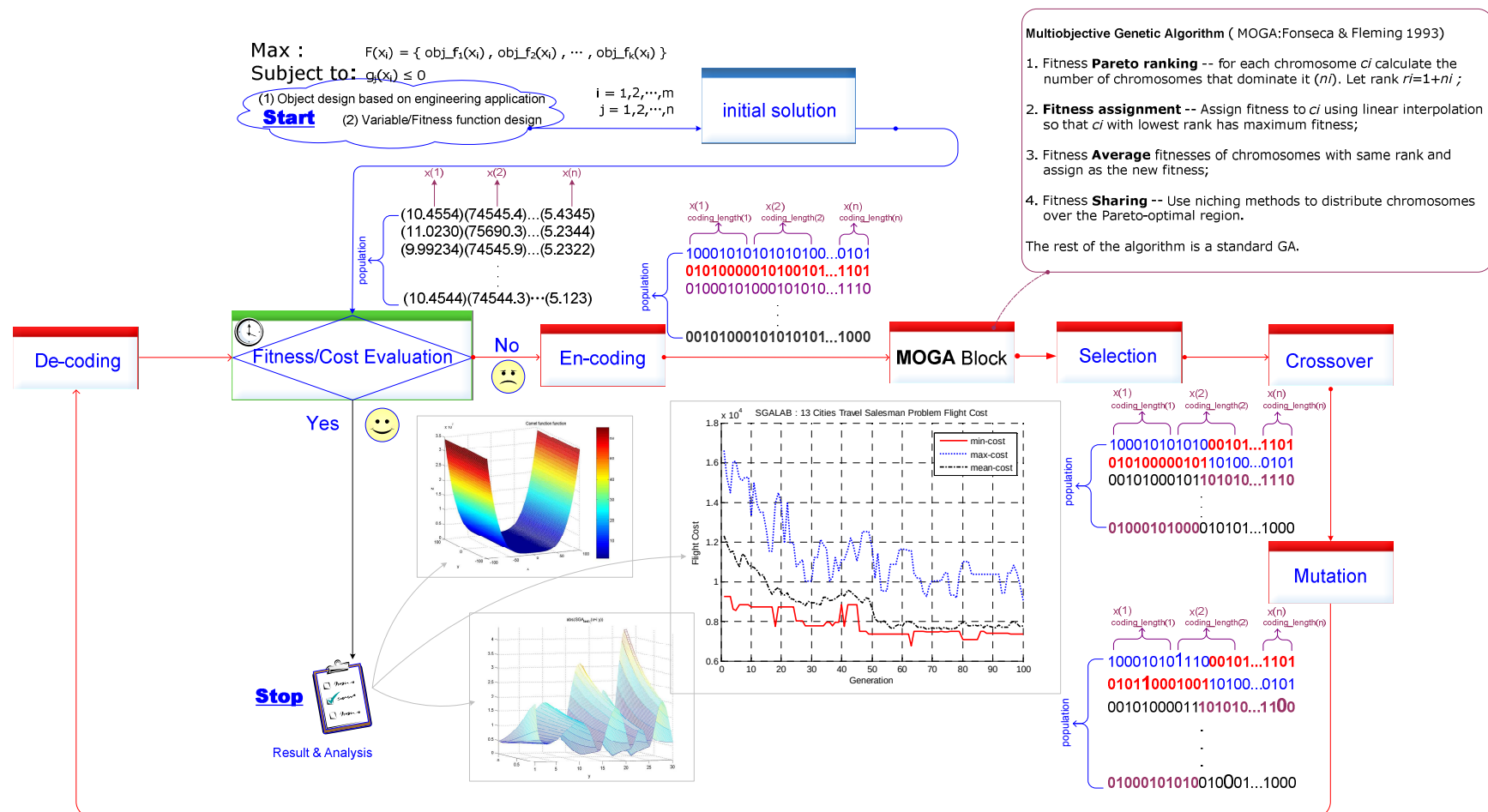
FAQ 3 – What is MOGA?

The Multiple Objective Genetic Algorithms (MOGA) was proposed by Fonseca and Fleming in 1993 [6],

Multiobjective Genetic Algorithm (MOGA)

Fonseca, C. M. & P. J. Fleming: "Genetic algorithms for multiobjective optimization: formulation, discussion and generalization", *Proc. of the 5th Inter. Conf. on GAs*, pp. 416-423, 1993

Chen Yi
chenyi2005@gmail.com
28th Nov., 2005



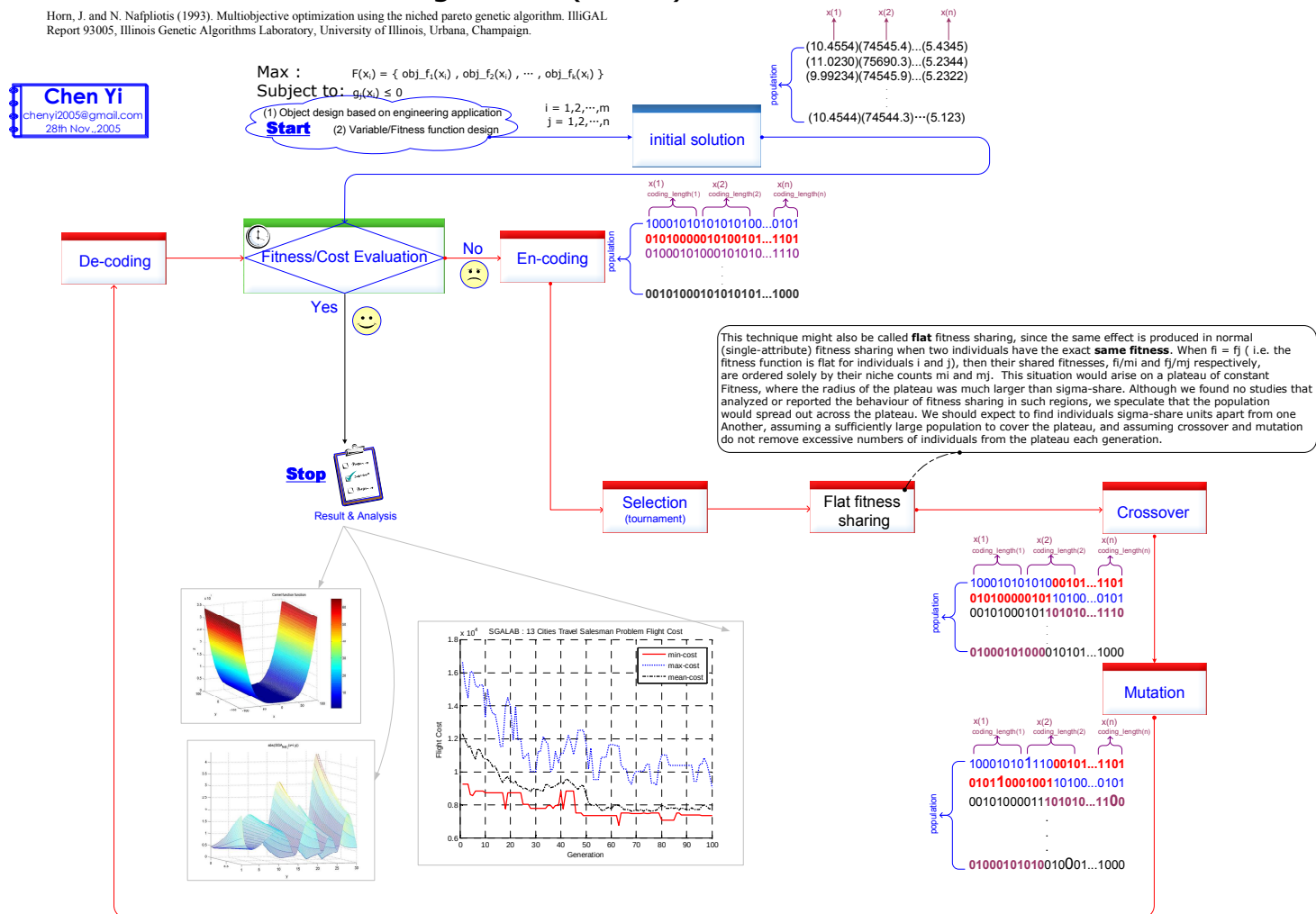
FAQ 4 – What is NPGA?

The Niche-Pareto Genetic Algorithm (NPGA) is proposed by Horn et al.[7].

Niched Pareto Genetic Algorithm (NPGA)

Horn, J. and N. Nafpliotis (1993). Multiobjective optimization using the niched pareto genetic algorithm. IlliGAL Report 93005, Illinois Genetic Algorithms Laboratory, University of Illinois, Urbana, Champaign.

Chen Yi
chenyi2005@gmail.com
28th Nov., 2005



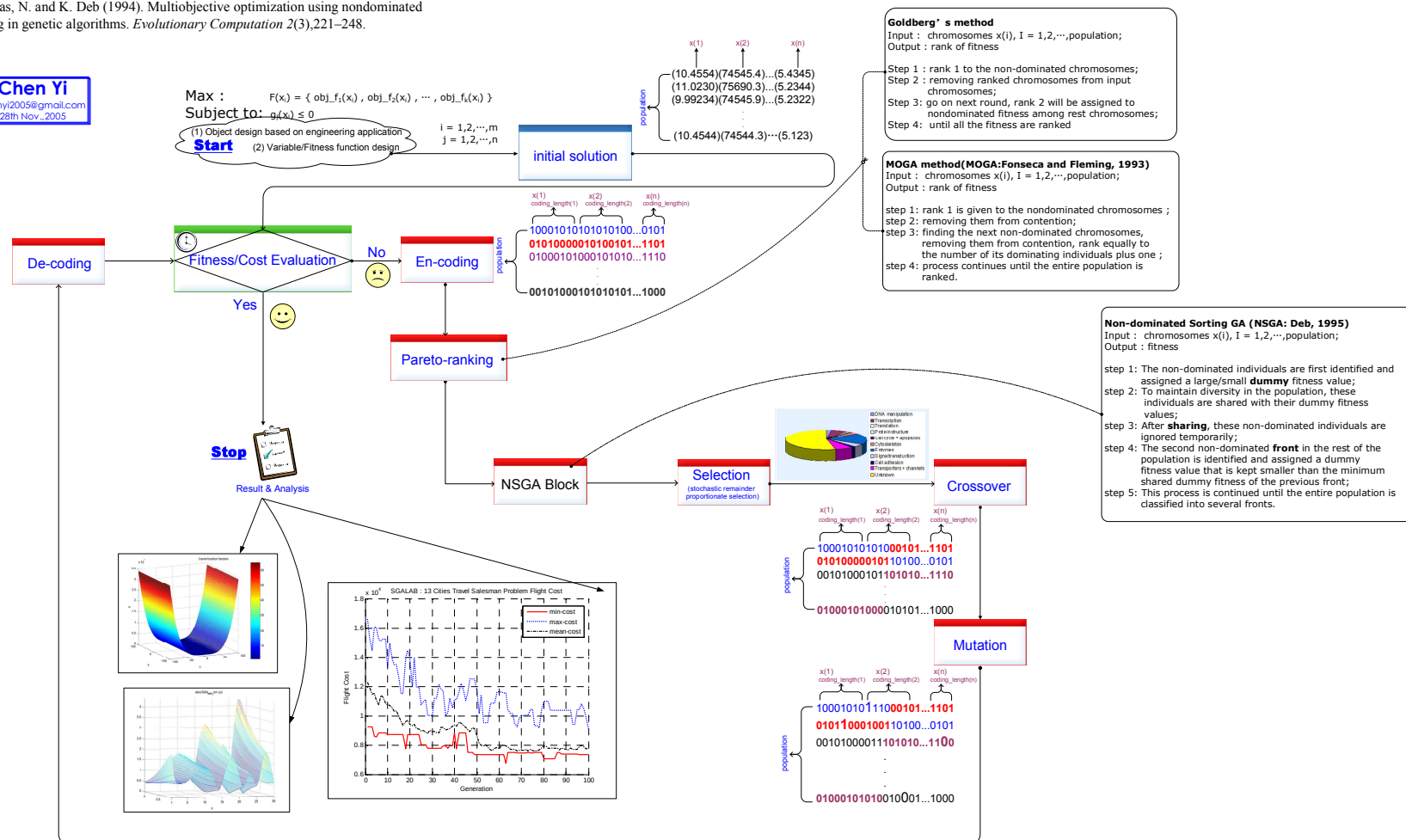
FAQ 5 – What is NSGA?

The Nondominated Sorting Genetic Algorithm (NSGA) is proposed by Srinivas and Deb [8].

Non-dominated Sorting Genetic Algorithm(NSGA)

Srinivas, N. and K. Deb (1994). Multiobjective optimization using nondominated sorting in genetic algorithms. *Evolutionary Computation* 2(3),221–248.

Chen Yi
chenyi2005@gmail.com
28th Nov., 2005



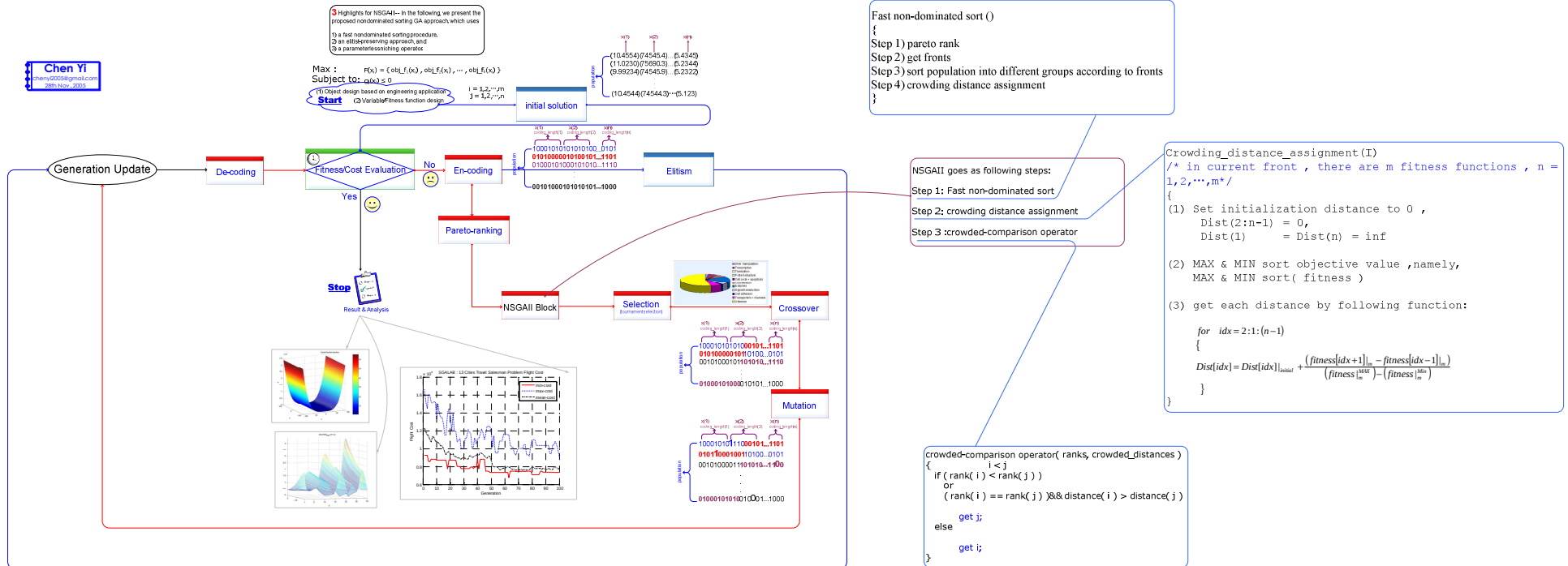
FAQ 6 – What is NSGAII?

The Nondominated Sorting Genetic Algorithm II (NSGAII) is proposed by Deb et al.[9],

Non-dominated Sorting Genetic Algorithm-II(NSGA-II)

[1] Deb, K.: *Multi-objective optimization Using Evolutionary Algorithms*, John Wiley, 2001.

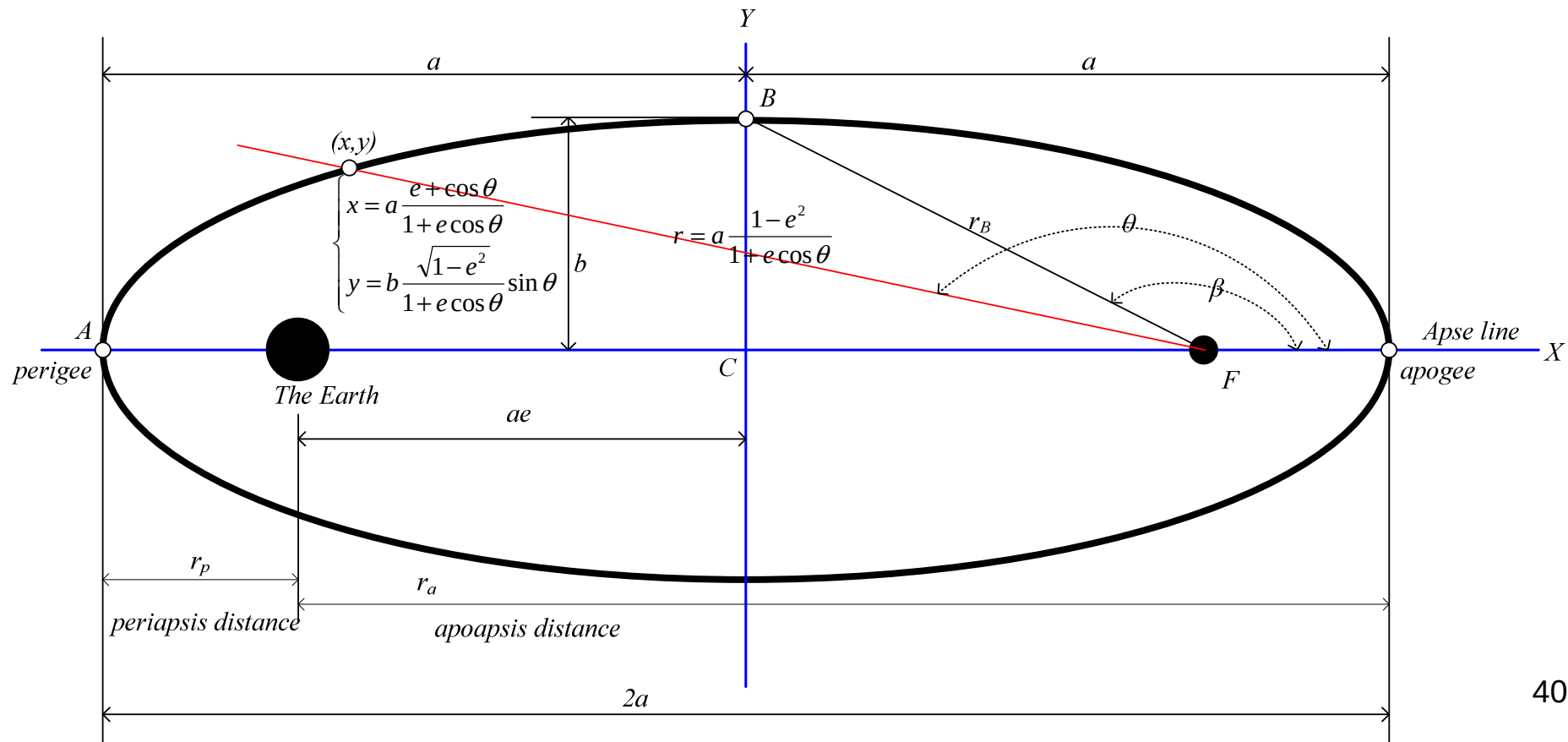
[2] Deb, K., A. Pratap, S. Agarwal and T. Meyarivan: "A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II", *IEEE Trans. on Evolutionary Comput.*, vol.6, no.2, 182-197, 2002.



FAQ 7 – Perigee & Apogee

Perigee is the point at which an object in orbit around the Earth makes its **closest** approach to the Earth.

Apogee is the point at which an object in orbit around the Earth makes its **greatest** approach to the Earth.

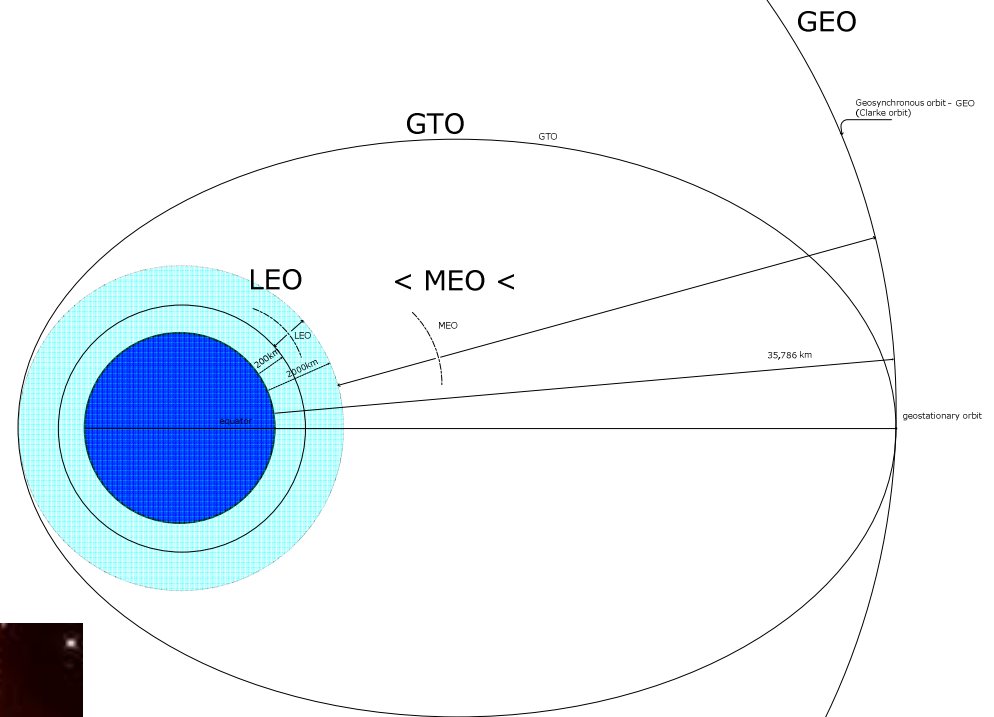
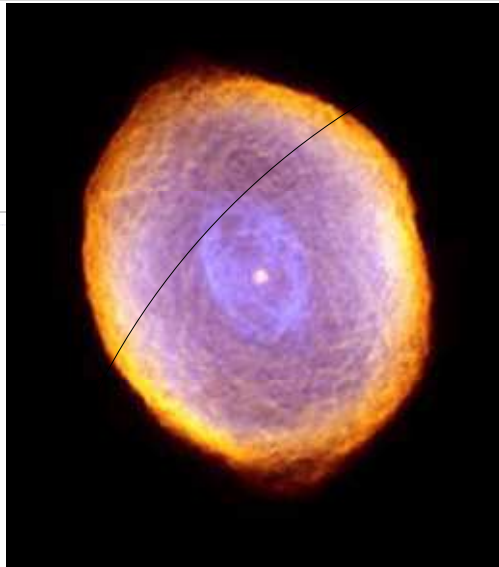


FAQ 8 – LEO

A **low Earth orbit** (LEO) is generally defined as an [orbit](#) within the [locus](#) extending from the Earth's surface up to an altitude of 2,000 km.

Higher orbits include [medium Earth orbit](#) (MEO), sometimes called [intermediate circular orbit](#) (ICO), and further above, [Geostationary orbit](#) (GEO). Orbits higher than low orbit can lead to earlier failure of electronic components due to intense radiation and charge accumulation, while commercial devices such as laptops have been used successfully in LEO during manned flight

http://en.wikipedia.org/wiki/Low_Earth_orbit



FAQ 9 – SGALAB

The Simple Genetic Algorithms Laboratory (SGALAB) toolbox for MATLAB is the GAs simulation tool for MMET payload transfer optimisation.

<http://www.mathworks.com/matlabcentral/fileexchange/loadFile.do?objectId=5882&objectType=file>



The screenshot shows the MATLAB Central File Exchange interface. At the top, the MATLAB Central logo is displayed with the tagline "An open exchange for the MATLAB and Simulink user community". A search bar contains the text "File Exchange". Below the logo, a navigation bar includes links for "File Exchange", "MATLAB Newsgroup", "Link Exchange", "Blogs", and "Contest". The main content area shows the breadcrumb path: "MATLAB Central > File Exchange > Optimization > SGALAB1003beta5.0.0.2(NSGA-II ParetoFront)". The title of the file is "SGALAB1003beta5.0.0.2(NSGA-II ParetoFront)". Below the title, the download format is ".zip". The rating is shown as four stars and a half, with "35 reviews" and a link to "Review this file". The author is "Yi Chen". The summary states "Genetic Algorithms Toolbox" and "MATLAB Release: R14SP2". The required products are "Communications Blockset". A description section lists updates for SGALAB 1003 Beta 5.0.0.2, including updates to the User's Guide, addition of Pareto Front plots, and updates for SGALAB 1003 Beta 5.0.0.1, including updates to the WhatisGAs, User's Guide, and QuickStart PDFs, and the addition of a passive suspension model for FLC and skyhook control. A small image of the toolbox packaging is shown with a link to "View fullsize image".

Download Now: .zip

Rating: ★★★★★ 35 reviews [Review this file](#)

Author: [Yi Chen](#)

Summary: Genetic Algorithms Toolbox

MATLAB Release: R14SP2

Required Products: Communications Blockset

Description:

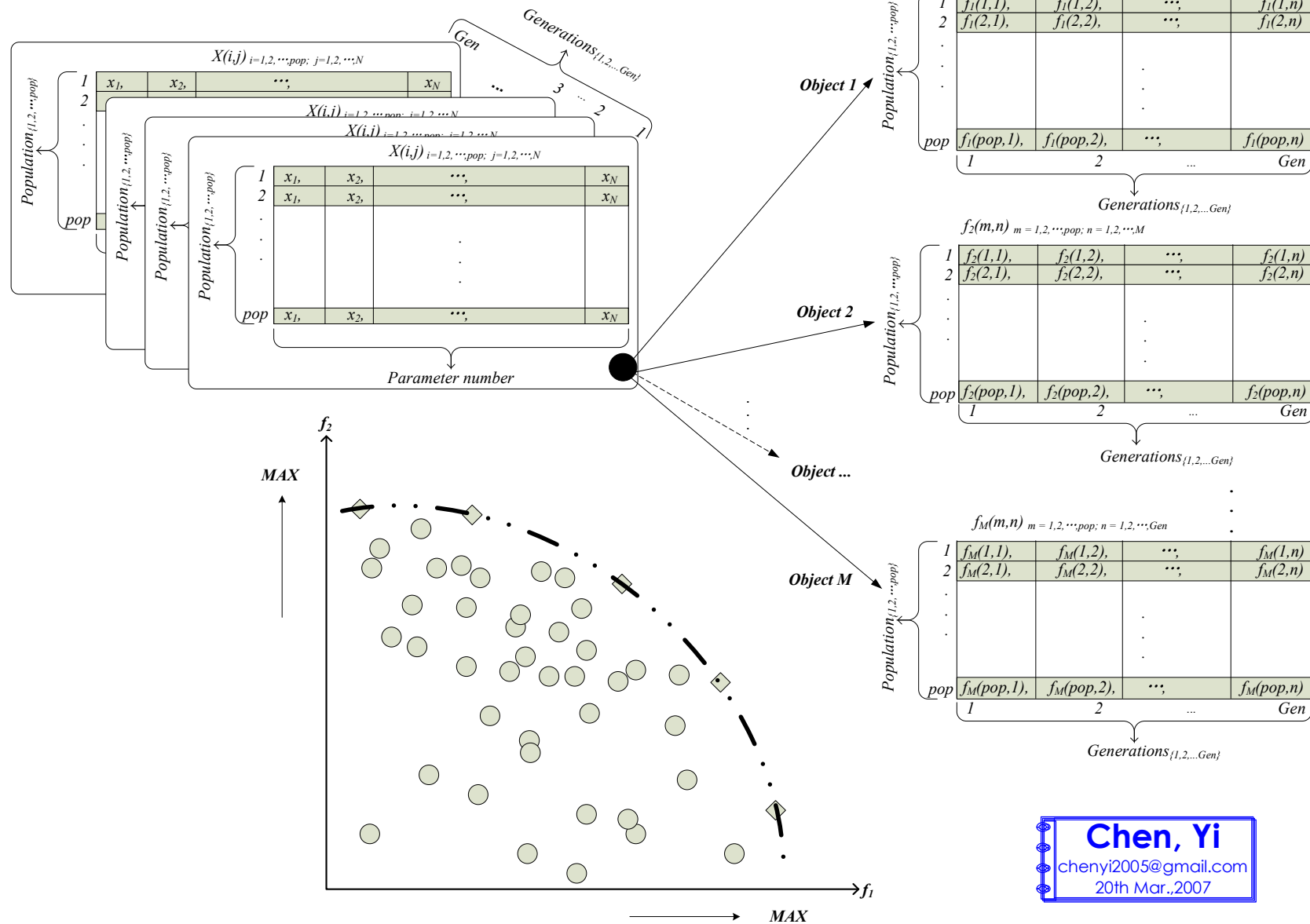
SGALAB 1003 Beta 5.0.0.2

- 1) Update SGALAB_FAQ_User's Guide.pdf
- 2) Add Pareto Front plots

SGALAB 1003 Beta 5.0.0.1

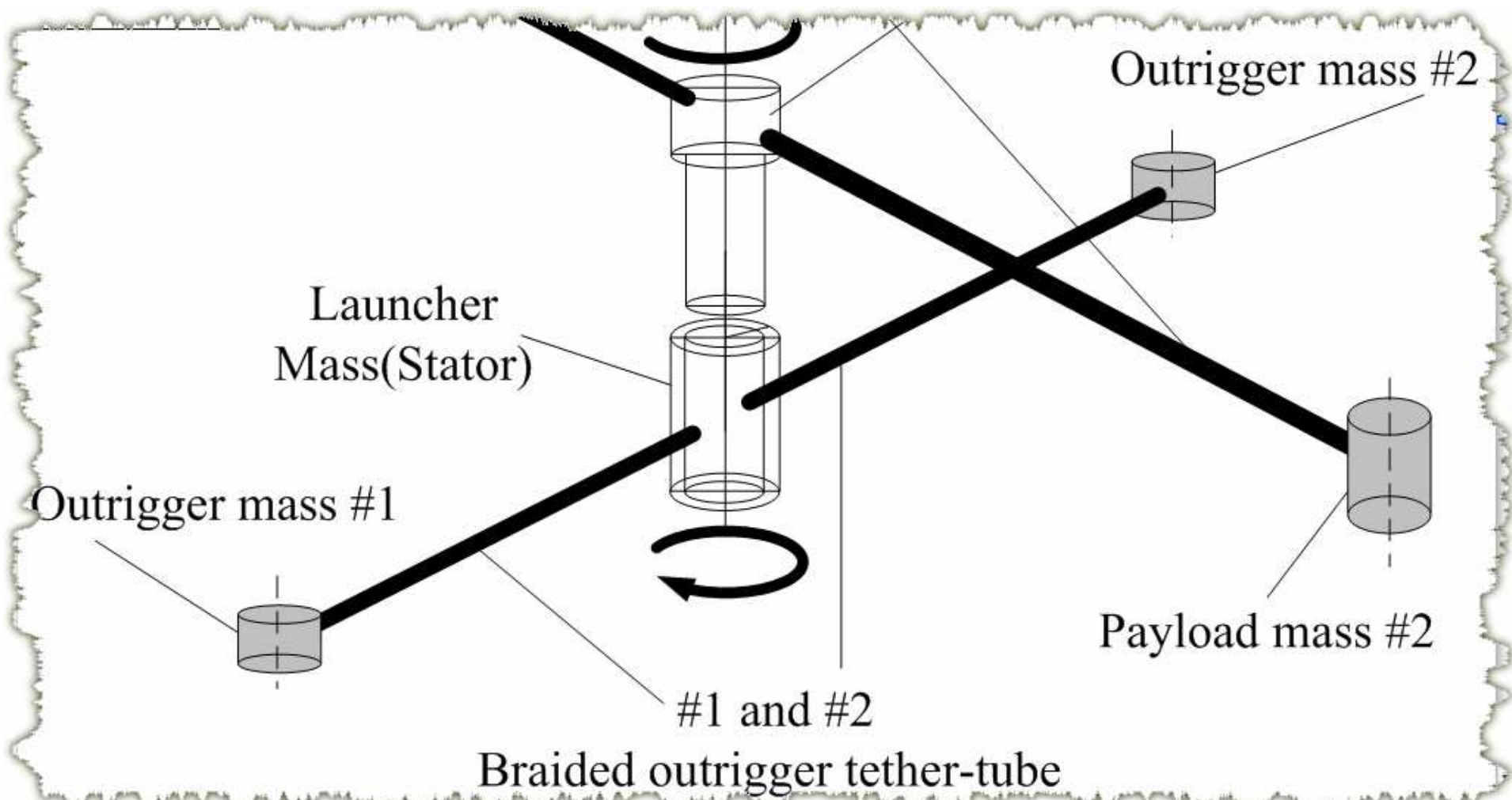
- 1) Update SGALAB_FAQ_WhatIsGAs.pdf
- 2) Update SGALAB_FAQ_User's Guide.pdf
- 3) Add SGALAB_FAQ_QuickStart.pdf
- 4) Add passive suspension model of FLC and skyhook, and GA+FLC and GA+Skyhook control will be available in SGALAB 1003 Beta 6

FAQ 9 – Pareto Front



FAQ 10 – Outrigger

structure on aircraft: a projection attached to an aircraft or other vehicle or machine to stabilize it or to support something



<code>\alpha</code>	α	<code>\upsilon</code>	υ	<code>\sim</code>	$\sim$
<code>\beta</code>	β	<code>\phi</code>	Φ	<code>\leq</code>	$\leq$
<code>\gamma</code>	γ	<code>\chi</code>	χ	<code>\infty</code>	∞
<code>\delta</code>	δ	<code>\psi</code>	Ψ	<code>\clubsuit</code>	$\clubsuit$
<code>\epsilon</code>	ϵ	<code>\omega</code>	ω	<code>\diamondsuit</code>	$\diamondsuit$
<code>\zeta</code>	ζ	<code>\Gamma</code>	Γ	<code>\heartsuit</code>	$\heartsuit$
<code>\eta</code>	η	<code>\Delta</code>	Δ	<code>\spadesuit</code>	$\spadesuit$
<code>\theta</code>	Θ	<code>\Theta</code>	Θ	<code>\leftrightarrow</code>	$\leftrightarrow$
<code>\vartheta</code>	ϑ	<code>\Lambda</code>	Λ	<code>\leftarrow</code>	$\leftarrow$
<code>\iota</code>	ι	<code>\Xi</code>	Ξ	<code>\uparrow</code>	$\uparrow$
<code>\kappa</code>	κ	<code>\Pi</code>	Π	<code>\rightarrow</code>	$\rightarrow$
<code>\lambda</code>	λ	<code>\Sigma</code>	Σ	<code>\downarrow</code>	$\downarrow$
<code>\mu</code>	μ	<code>\Upsilon</code>	Υ	<code>\circ</code>	$\circ$
<code>\nu</code>	ν	<code>\Phi</code>	Φ	<code>\pm</code>	$\pm$
<code>\xi</code>	ξ	<code>\Psi</code>	Ψ	<code>\geq</code>	$\geq$
<code>\pi</code>	π	<code>\Omega</code>	Ω	<code>\propto</code>	$\propto$
<code>\rho</code>	ρ	<code>\forall</code>	$\forall$	<code>\partial</code>	∂
<code>\sigma</code>	σ	<code>\exists</code>	$\exists$	<code>\bullet</code>	$\bullet$
<code>\varsigma</code>	ς	<code>\ni</code>	$\ni$	<code>\div</code>	$\div$
<code>\tau</code>	τ	<code>\cong</code>	$\cong$	<code>\neq</code>	$\neq$
<code>\equiv</code>	$\equiv$	<code>\approx</code>	$\approx$	<code>\aleph</code>	$\aleph$
<code>\Im</code>	$\Im$	<code>\Re</code>	$\Re$	<code>\wp</code>	$\wp$
<code>\otimes</code>	$\otimes$	<code>\oplus</code>	$\oplus$	<code>\oslash</code>	$\oslash$
<code>\cap</code>	$\cap$	<code>\cup</code>	$\cup$	<code>\supseteq</code>	$\supseteq$
<code>\supset</code>	$\supset$	<code>\subseteq</code>	$\subseteq$	<code>\subset</code>	$\subset$
<code>\int</code>	$\int$	<code>\in</code>	$\in$	<code>\o</code>	$\circ$
<code>\rfloor</code>	$\rfloor$	<code>\lceil</code>	$\lceil$	<code>\nabla</code>	∇
<code>\lfloor</code>	$\lfloor$	<code>\cdot</code>	$\cdot$	<code>\ldots</code>	$\ldots$
<code>\perp</code>	$\perp$	<code>\neg</code>	$\neg$	<code>\prime</code>	$\prime$

[西江月.秋]

小轩窗秋叶入，
穹庐间月影疏。
轻寒烟岚老鸦树，
只雁影瘦山湖。

野草带袂孤途，
琵琶夜语霜露。
古今成败笑谈故，
渔樵闲话出处。

Chen, Yi
leo.chen.yi@gmail.com

Room429 James Watt South Building,
University of Glasgow.
05-Sep-2007, 有风雨, 今天特别冷